

MALAHAT, BC, Canada

WMO: 717740

Lat: 48.575N

Lon: 123.530W

Elev: 366

StdP: 97.01

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.3	-3.3	-12.7	1.3	3.0	-10.6	1.6	5.0	7.4	2.9	6.7	2.7	3.2	30	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.6	27.7	16.8	25.8	16.3	24.1	15.8	18.8	24.6	17.6	23.7	16.6	22.8	1.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	12.3	22.0	15.1	11.2	20.4	14.0	10.4	19.1	54.9	24.4	50.7	23.8	47.8	22.8	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.7	4.9	-7.2	31.0	2.7	1.7	-9.1	32.2	-10.7	33.2	-12.2	34.1	-14.2	35.3		
(5)			-7.4	19.1	2.3	2.1	-9.1	20.7	-10.4	21.9	-11.7	23.1	-13.3	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	3.5	3.6	5.4	8.0	11.9	14.7	18.2	18.4	15.0	9.6	5.4	3.0
	DBStd	6.33	3.09	2.87	2.73	2.99	3.61	3.48	3.39	3.09	3.21	2.63	2.98	2.80
	HDD10.0	1021	202	179	146	76	19	2	0	0	1	39	140	217
	HDD18.3	3233	460	413	402	309	201	121	44	38	111	271	388	476
	CDD10.0	932	0	0	3	16	79	142	255	259	150	27	2	0
	CDD18.3	104	0	0	0	0	3	12	41	39	10	0	0	0
	CDH23.3	537	0	0	0	0	9	71	237	188	33	0	0	0
	CDH26.7	101	0	0	0	0	1	13	56	28	3	0	0	0
(14) Wind	WSAvg	2.0	2.2	2.2	2.3	2.2	2.0	2.1	1.8	1.7	1.8	2.0	2.2	2.3
(15) Precipitation	PrecAvg	1292	218	137	120	69	43	35	22	33	45	139	223	214
	PrecMax	1694	370	371	256	146	86	88	49	121	126	370	473	321
	PrecMin	923	85	30	29	18	10	7	3	3	4	25	86	111
	PrecStd	216	75	71	49	32	20	17	13	29	32	75	101	63
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	10.6	15.8	20.2	24.5	28.5	31.2	29.3	26.5	18.5	12.8	10.5
		MCWB	8.4	7.7	8.8	10.8	14.7	15.8	19.2	16.9	15.2	11.4	10.7	9.3
	2%	DB	9.2	9.1	12.3	17.0	21.9	25.3	28.1	27.2	24.0	16.2	11.6	8.8
		MCWB	7.4	6.7	7.9	9.7	13.4	15.4	17.3	16.9	15.3	11.3	10.1	7.3
	5%	DB	8.2	8.1	10.6	14.7	19.9	22.7	25.8	25.5	21.9	14.6	10.3	7.5
		MCWB	6.4	6.1	7.1	8.9	12.6	14.3	16.7	16.5	14.3	10.5	9.0	6.2
	10%	DB	7.2	7.1	9.2	12.6	17.8	20.5	24.0	23.7	19.9	13.2	9.2	6.4
		MCWB	5.5	5.1	6.3	7.6	11.6	13.4	15.9	16.0	13.7	9.7	7.7	5.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	9.2	10.7	12.9	16.0	17.8	22.1	22.8	18.1	14.4	11.7	9.1
		MCDB	10.3	10.2	14.2	18.3	22.2	25.1	26.6	23.8	23.3	16.8	12.5	9.8
	2%	WB	8.0	7.7	9.1	11.2	14.6	16.5	19.3	18.9	16.7	12.9	10.5	7.6
		MCDB	9.0	8.8	11.6	15.9	20.0	23.6	25.3	24.4	22.0	14.7	11.4	8.3
	5%	WB	6.9	6.7	7.9	9.8	13.6	15.3	17.7	17.6	15.7	11.6	9.5	6.4
		MCDB	8.0	7.9	10.2	13.6	18.5	21.5	23.6	23.7	20.6	13.7	10.5	7.1
	10%	WB	5.9	5.8	7.0	8.6	12.5	14.3	16.6	16.5	14.7	10.5	8.2	5.4
		MCDB	7.0	6.8	9.2	12.0	17.3	19.7	22.7	22.7	19.0	12.6	9.2	6.2
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	4.2	5.4	6.7	7.6	8.1	9.0	8.6	7.0	5.0	3.8	3.1
		MCDBR	4.2	5.1	6.9	9.2	9.9	10.8	10.9	10.4	9.2	6.9	4.6	4.2
	5% WB	MCWBR	3.7	3.8	4.7	5.6	6.0	5.9	5.6	5.3	5.1	4.5	3.9	3.7
		MCWBR	4.0	4.3	6.3	8.1	8.9	9.6	10.1	9.5	8.1	6.0	4.4	3.9
(40) Clear-Sky Solar Irradiance	taub		0.303	0.300	0.315	0.341	0.357	0.339	0.331	0.335	0.322	0.316	0.308	0.298
		taud	2.559	2.589	2.560	2.474	2.445	2.507	2.570	2.553	2.589	2.590	2.552	2.536
	Ebn at Noon		781	873	910	912	904	921	924	907	891	839	765	737
		Edh at Noon	58	70	84	101	108	102	95	92	81	69	58	53
(44) All-Sky Solar Radiation	RadAvg		0.92	1.80	2.81	4.27	5.47	5.97	6.52	5.50	3.91	2.15	1.10	0.74
	RadStd		0.12	0.24	0.33	0.40	0.43	0.57	0.46	0.36	0.32	0.24	0.17	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page